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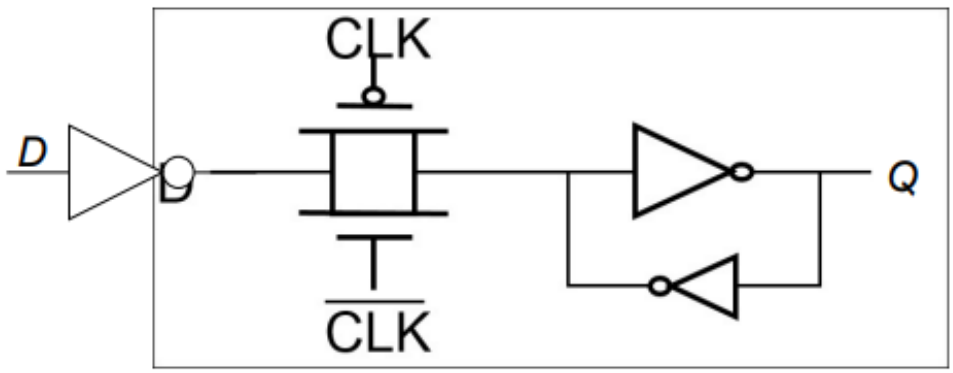
1. Part 1 – Latch with Weak Feedback Inverter

1.1 Introduction

In this part, a transistor-level implementation of a latch with weak inverter feedback is designed and simulated in ngspice to verify its functional behavior.



The expected logical behavior of circuit above is described in the truth table below. The node after the transmission gate will be named "X":

D	CLK	X	Q
0	0	1	0
0	1	Previous Value	Previous Value
1	0	0	1
1	1	Previous Value	Previous Value

When the clock is low, the output Q follows the input D, while for a high clock level the previously stored value is retained.

1.2 Implementation/Verification

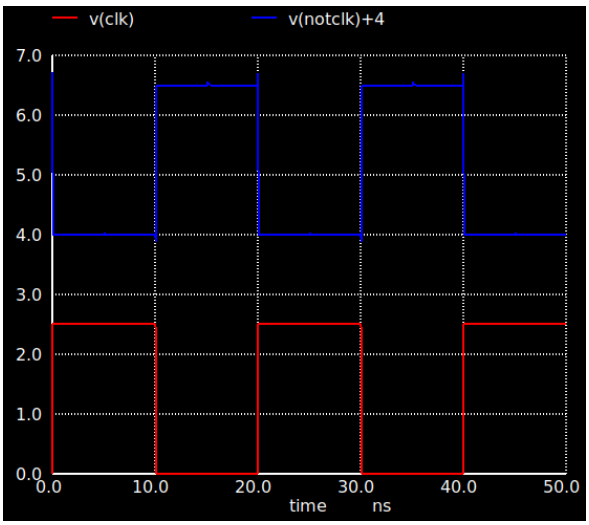
The weak feedback inverter was implemented using the minimum allowed transistor sizes:

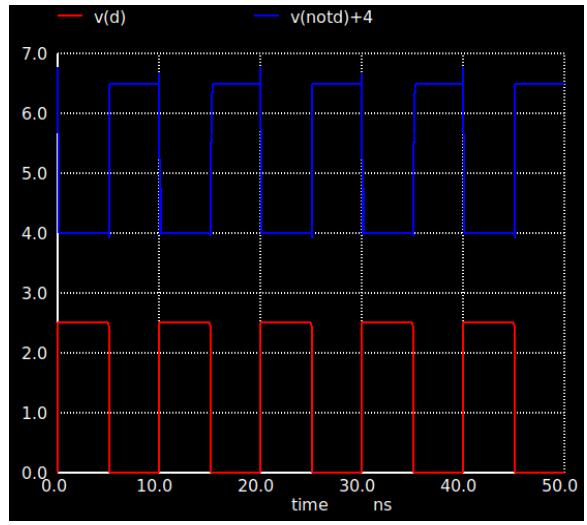
$$Wn = 3\ \mu m,\ Wp = 9\ \mu m,$$

while the remaining inverters and the transmission gate were sized ×2:

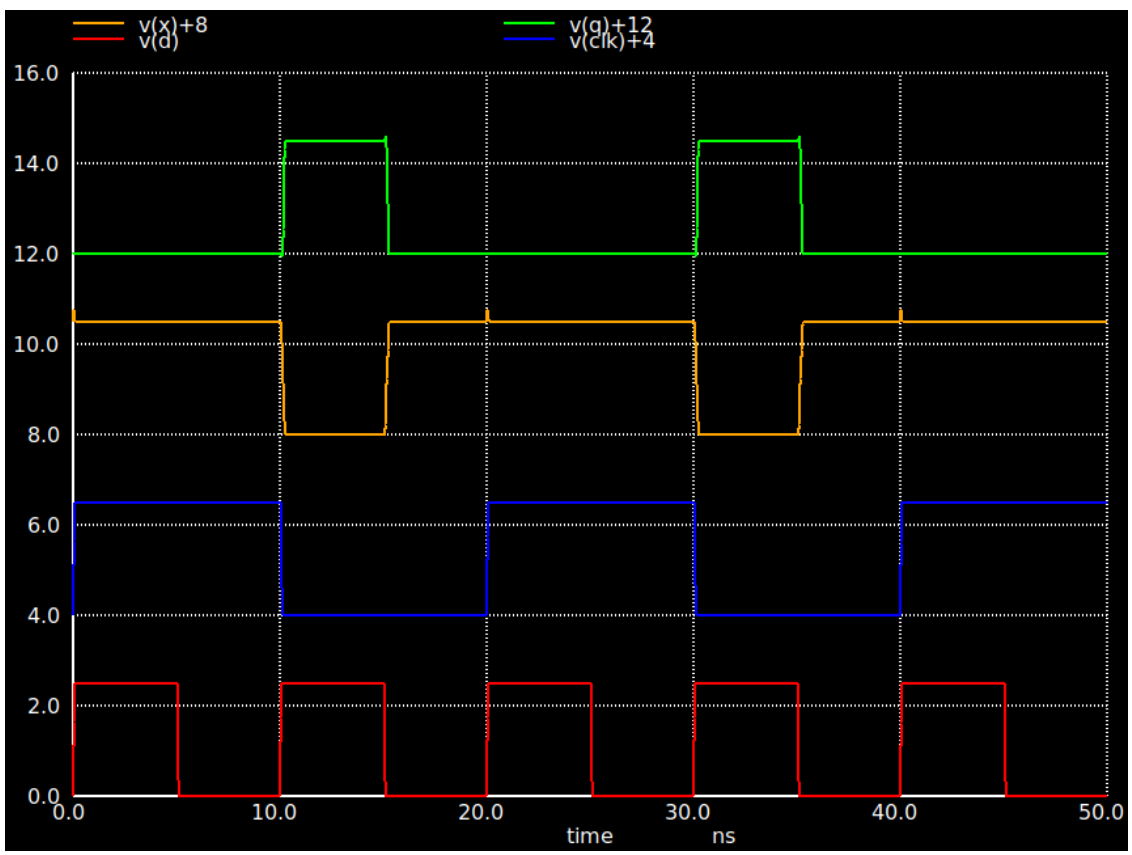
$$Wn = 6\ \mu m, Wp = 18\ \mu m$$

The goal of the verification is to show that the circuit behaves the same way as the truth table shown above. Firstly, the D' and CLK' signals were checked, and as seen in the waveforms below they are produced correctly.





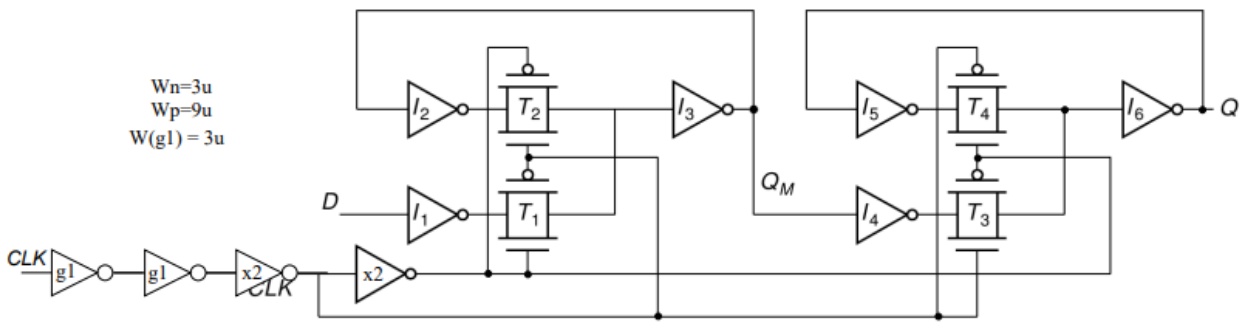
Moving to the full waveform that shows the complete transient response of the latch, including D, CLK, the internal node X and the output Q. The waveform confirms that Q follows D when the transmission gate is transparent (CLK = 0) and remains the same when the transmission gate is closed, in agreement with the truth table. Therefore, the circuit is designed correctly.



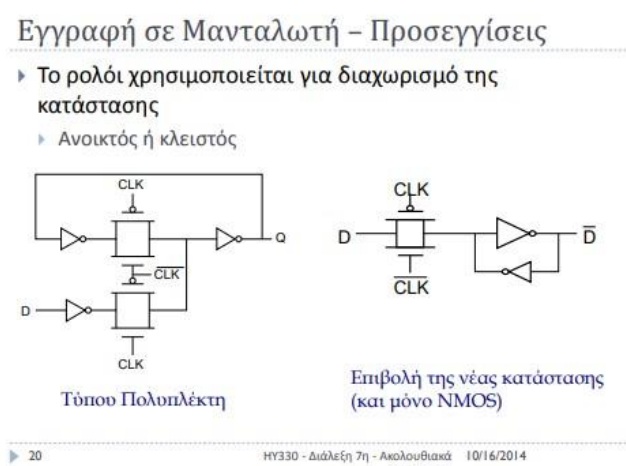
2. Part 2 – Master-Slave Flip-Flop

2.1 Introduction

In this part, a transistor-level implementation of a Master-Slave Flip-Flop is designed and simulated in ngspice to verify its functional behavior.



I decided to modify the above schematic provided in the assignment. I did this with the help of a transformation presented in the lecture notes:



This transformation is allowing the reuse of the latch designed and verified in Part A, which also uses less transistors. The only functional difference introduced by this approach is that the output Q no longer follows the inverted input, since no inverter is placed at the input D, in accordance with the lecture description. But in this implementation where there are a master and a slave, the output of the master Qm will be D' and the output of the slave Q will be Qm'.

The transformation will result in Q following D as intended, with the only difference that Qm is going to be the opposite of D.

The expected logical behavior of the circuit above is described in the truth table below. The node after the master's transmission gate will be named "XM", while the node after the slave's transmission gate will be named "XS":

D	CLK	XM	QM	XS	Q
0	0	0	1	Previous Value	Previous Value
0	1	Previous Value	Previous Value	1	0
1	0	1	0	Previous Value	Previous Value
1	1	Previous Value	Previous Value	0	1

2.2 Implementation/Verification

2.2.1 Part A

The weak feedback inverters and the two-unit clock buffer inverters (g1) were implemented using the minimum transistor dimensions

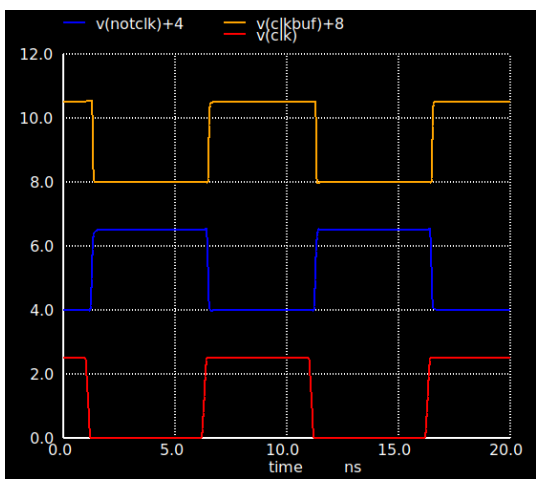
$$Wn = 3\ \mu m,\ Wp = 9\ \mu m,$$

All remaining inverters and the transmission gate were sized ×2:

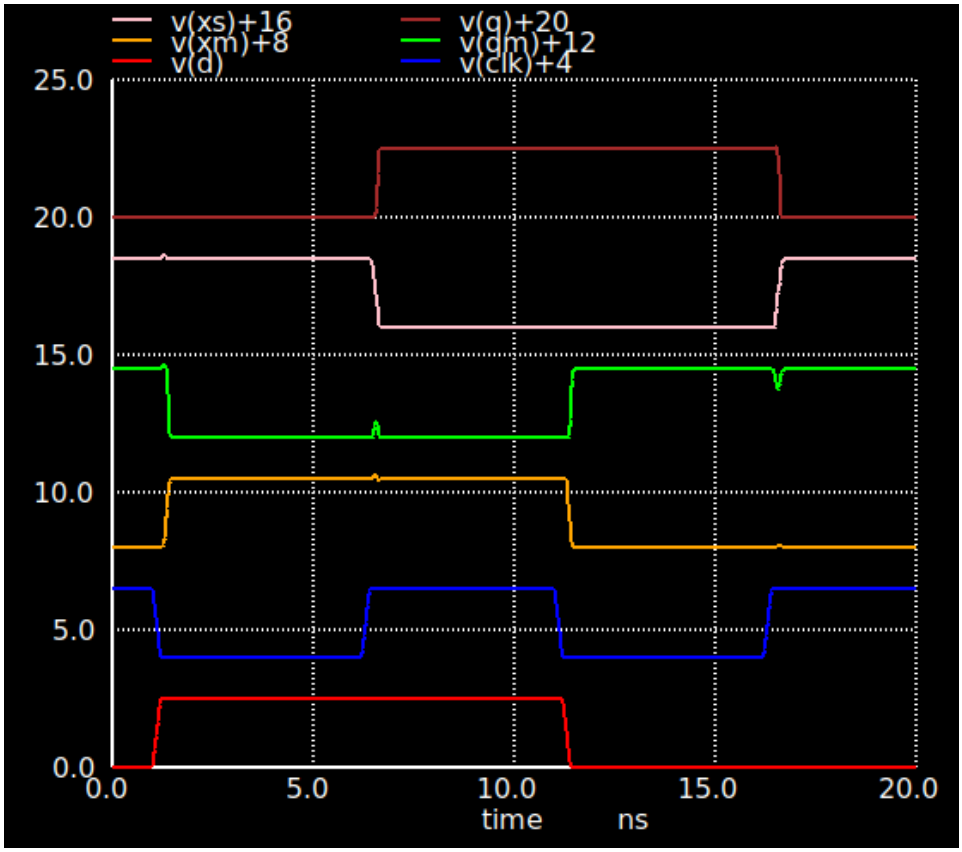
$$Wn = 6\ \mu m, Wp = 18\ \mu m$$

The master latch transmission gate is transparent when CLK = 0, while the slave latch transmission gate is transparent when CLK = 1.

Now again starting the verification process, the CLK' and the CLKbuf (which is the CLK signal after passing the 4 inverters at the start) signals were first checked, and as seen in the waveform below they are produced correctly.



Moving to the full waveform that shows the complete transient response of the Flip-Flop, including D, CLK, the internal nodes XM and XS QM and the output Q. The waveform shows that QM updates to the opposite of D when CLK = 0, while the Q stays constant. while when CLK = 1 The Q updates to the opposite of QM (meaning D) while the QM stays constant. The internal nodes XM and XS are always the opposites of QM and Q respectively. Therefore, the circuit is designed correctly, proven also by the truth table.

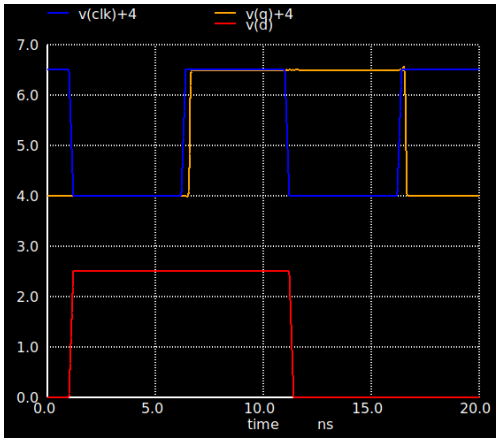


2.2.2 Part B

Measurements were performed for two different input rise/fall times at D (200 ps and 400 ps) and for two output loading conditions (CQ = Cg1 and CQ = 2Cg1), while keeping the clock rise/fall time fixed at 200 ps, as required.

2.2.2.1 CLK → Q Propagation Delay

The CLK→Q propagation delay was measured using transient simulations by detecting the 50% voltage crossing points (1.25 V) of the clock and the output Q. Both low-to-high and high-to-low transitions were evaluated using ngspice ".measure" commands. An indicative waveform follows:



The Results

For a D rise/fall time of 200ps and 400ps with a Cg1 = 1fF:

- Time CLK → Q low to high = 2.964129e-10 s
- Time CLK → Q high to low = 3.043960e-10 s

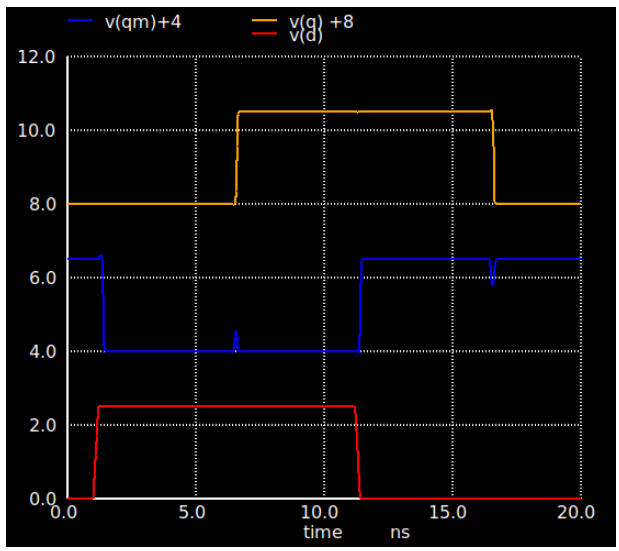
For a D rise/fall time of 200ps and 400ps with a Cg1 = 2fF:

- Time CLK → Q low to high = 2.969665e-10 s
- Time CLK → Q high to low = 3.050640e-10 s

The rise and fall times of D didn't make any difference to the time propagation.

2.2.2.2 Rise and Fall Times of QM and Q

The rise and fall times of the internal node QM and the output Q were measured between the 10% and 90% voltage levels (0.25 V to 2.25 V) using transient ".measure" commands. An indicative waveform follows:



The Results

For a D rise/fall time of 200ps and with a Cg1 = 1fF:

- Rise time of QM = 4.933977e-09 s
- Fall time of QM = 5.826000e-11 s
- Rise time of Q = 7.024504e-11 s
- Fall time of Q = 7.677451e-11 s

For a D rise/fall time of 400ps and with a Cg1 = 1fF:

- Rise time of QM = 5.253720e-09 s
- Fall time of QM = 5.898566e-11 s
- Rise time of Q = 7.024504e-11 s
- Fall time of Q = 7.677451e-11 s

For a D rise/fall time of 200ps and with a Cg1 = 2fF:

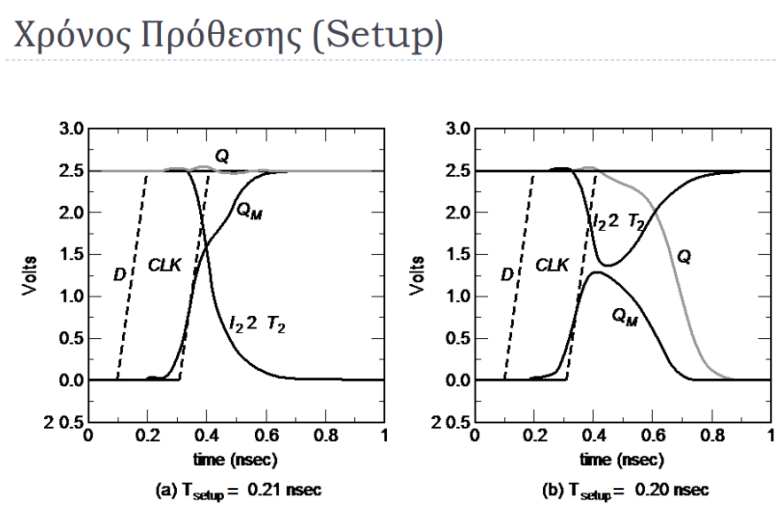
- Rise time of QM = 4.933977e-09 s
- Fall time of QM = 5.825999e-11 s
- Rise time of Q = 7.096511e-11 s
- Fall time of Q = 7.735610e-11 s

For a D rise/fall time of 400ps and with a Cg1 = 2fF:

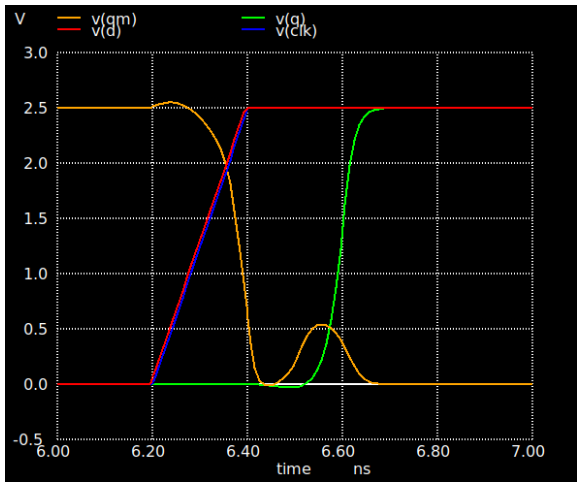
- Rise time of QM = 5.253720e-09 s
- Fall time of QM = 5.898564e-11 s
- Rise time of Q = 7.096511e-11 s
- Fall time of Q = 7.735610e-11 s

2.2.2.3 Setup Limit

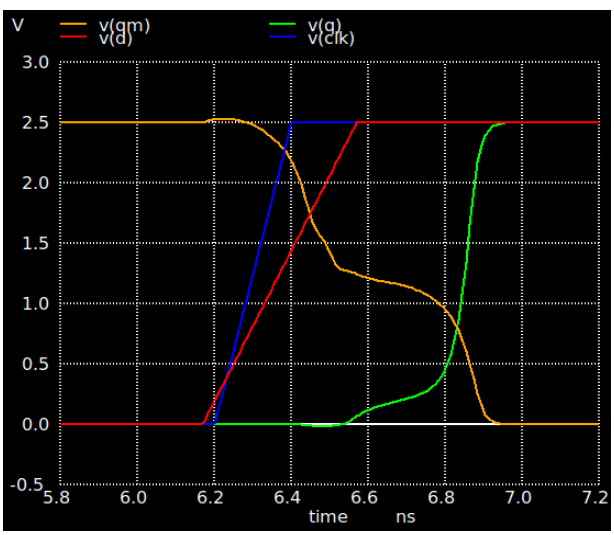
The setup time was investigated by shifting the input D close to CLK, so that if the rises happen very close it would cause a metastability problem in the output as described in the lecture notes:

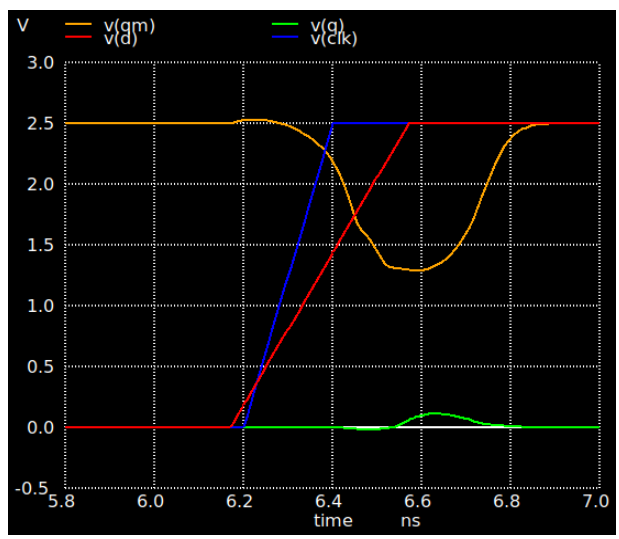


Despite extensive simulations, true metastability could not be caused in the implemented circuit, and therefore a clear setup time limit could not be extracted. Even with 0.001n distance between the D and CLK the circuit was working. Even without any distance the circuit was fine:



The only observable degradation in behavior occurred when the rise time of D was increased to 400 ps, exceeding the CLK rise time of 200 ps, and when the input D transition was shifted extremely close to the CLK so that the rising peak of D would extend beyond the CLK rising peak transition. Below the right waveform shows metastability while in the left the D starts rising just far enough of the CLK so that the circuit would be fine.



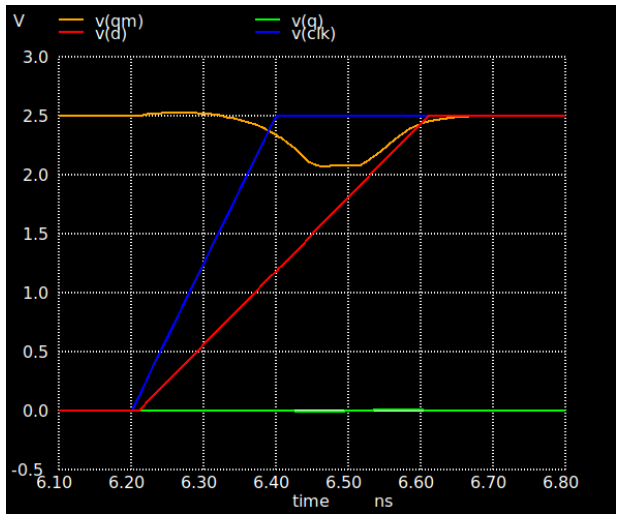
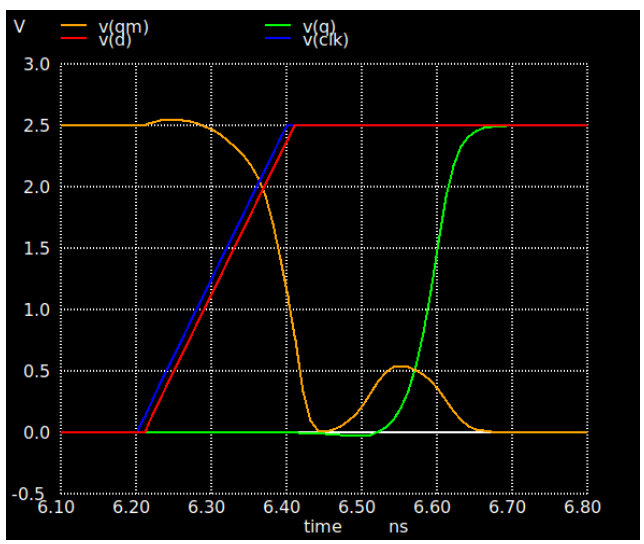


In the left picture the D and CLK start rising with a difference of 0.31ns and while the right picture with a difference of 0.30ns.

These results indicate that, due to the strong regenerative feedback and the sizing of the latch, the circuit prevents the occurrence of metastability.

2.2.2.4 Hold Limit

The hold time was evaluated by shifting the input D transition to occur after the active clock edge, while monitoring the output Q and the internal node QM. During the simulations, the input D was moved progressively closer to the clock edge from the positive time side. For a D rise time of 400ps the output Q remained unaffected by changes in D immediately after the clock transition.



But for a D rise time of 200ps the output Q followed the input's D value.